Digital Forensics 1
Rubric 2

Evidence Collection & Preservation
The project must include successful capture of live data from a running system using FTK Imager or KAPE. Create a verified disk image with MD5/SHA-1 hash validation. Acquire system memory using DumpIt or similar tools. Document the entire collection process with timestamps, tool versions, command parameters, and hash verification. All evidence must use write-blocking mechanisms to preserve integrity.

To successfully satisfy the **Evidence Collection & Preservation** section of this rubric, I need to execute and document a strictly controlled, forensically sound acquisition process.

All evidence collection must prioritize the Order of Volatility (RAM first, then the disk image) and utilize a write-blocking mechanism where applicable to preserve data integrity.

Here is the step-by-step breakdown of how to execute these tasks and document them perfectly for your rubric.

🛡️ 1. Core Constraint: Write-Blocking

Before touching any evidence, I must establish that write-blocking is active to prevent the host OS from modifying the evidence drive.

- **Hardware:** Note the use of a hardware write-blocker (e.g., Tableau, CRU WiebeTech) if in a physical lab environment.
- **Software:** If operating in a virtual or purely software environment, explicitly document the registry modification to enable software write-blocking:
    - **Path:** `HKLM\SYSTEM\CurrentControlSet\Control\StorageDevicePolicies`
    - **Value:** `WriteProtect` set to `1` (DWORD).

- **Documentation Requirement:** Take a screenshot of your write-blocker configuration or the registry key before starting collections.

🧠 2. Volatile Data Capture: System Memory (RAM)

Because system memory changes constantly, you must acquire RAM **before** creating a disk image to satisfy the order of volatility.

- **Tool Choice:** Use **DumpIt** (part of the Comae toolkit) or **FTK Imager** (Capture Memory feature).
- **Execution (DumpIt Example):**
  - Run the command prompt as an Administrator from your trusted forensic external drive.
  - Execute the tool using the required parameters.
- **Required Documentation Elements:**
  - **Timestamp:** Exact date and time execution started and finished.
  - **Tool Version:** e.g., `DumpIt v3.0.20170620`.
  - **Command Parameters:** `DumpIt.exe /OUTPUT F:\Evidence\mem.raw /QUIET` (or your specific command).
  - **Screenshot:** Capture the command window running or completing the tool showing the output path.

💾 3. Live Data Capture & Disk Imaging

Next, you will capture a verified disk image of the target system storage.

- **Tool Choice: FTK Imager** (Standard for creating physical/logical images) or **KAPE** (Kroll Artifact Parser and Extractor, if doing a targeted live artifact triage collection).
- **Execution (FTK Imager Example):**
  - File → **Create Disk Image**.

    - Select **Physical Drive** or **Logical Drive** (depending on your prompt guidelines; Physical is preferred for full forensic analysis).
    - Choose **Live System** drive as the source.
    - Set the destination image type to **E01 (Expert Witness Format)** or **RAW/DD**. *E01 is highly recommended as it embeds the hashes and metadata directly into the evidence file.*
    - Check the box for "**Verify images after they are created**".
- **Required Documentation Elements:**
    - **Timestamps:** Start time of image creation and verification completion time.
    - **Tool Version:** e.g., `AccessData FTK Imager v4.7.0.7`.
    - **Evidence Metadata:** Case Number, Evidence Number, Examiner Name, and Unique Description entered into the wizard.

🔑 4. Hash Validation & Verification

Your rubric explicitly requires a **verified disk image with MD5/SHA-1 hash validation**.

- **How to Capture It:** Once FTK Imager finishes the verification phase, it will display a "Drive Verification Results" window.
- **Screenshot Requirement:** You **must** take a screenshot of this specific window. It provides the ultimate proof of data integrity by listing:
    - **Computed Hash:** The hash calculated across the live disk.
    - **Reported Hash:** The hash calculated across the newly created image file.
    - **Match Result:** An explicit confirmation showing `"Match"` or `"Verified"`.
- **Text log:** Copy and paste the resulting `[Image_Name].txt` summary log file directly into your report appendix.

📝 Recommended Documentation Layout for Your Report

Group your findings under clear headings using a structured format like this for each asset collected:

Collection Metric	Memory Capture (RAM)	Disk Image (Live Capture)
Target Asset	System RAM (16GB)	Primary OS Partition (C:)
Tool Used & Version	DumpIt v3.0	FTK Imager v4.7.0
Command / Parameters	`DumpIt.exe /OUTPUT F:\Evidence\`	Physical Drive 0 → E01 Format
Execution Start Time	YYYY-MM-DD HH:MM:SS UTC	YYYY-MM-DD HH:MM:SS UTC
Execution End Time	YYYY-MM-DD HH:MM:SS UTC	YYYY-MM-DD HH:MM:SS UTC
Write-Blocker Info	Software Registry Block (`WriteProtect=1`)	Software Registry Block (`WriteProtect=1`)
MD5 Verification Hash	*[Insert 32-character string]*	*[Insert 32-character string]*
SHA-1 Verification Hash	*[Insert 40-character string]*	*[Insert 40-character string]*
Verification Status	Verified via internal tool log	**MATCH** (See Figure X)

Are you currently running these tools in a **provided virtual lab environment**, or are you setting up these configurations on a **local machine/custom VM**? I can give you the exact command strings or GUI click-paths depending on your setup.